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Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 6

1. What is potential real GDP?

Potential real GDP is

- ☐ A. the level of GDP attained when only some firms have excess capacity.
- ☒ B. the level of GDP attained when all firms are producing at capacity.
- ☐ C. the level of GDP attained when most firms are producing at capacity and unemployment is low.
- ☐ D. the level of GDP attained when all firms have excess capacity.

Potential real GDP grows over time .

2. Use the table below to answer the following questions.

Year	RGDP (billions of 2005 dollars)
1994	\$8,871
1995	8,998
1996	9,087
1997	9,890
1998	9,100

a. Calculate the growth rate of real GDP for each year from 1994 to 1998.

The growth rate of real GDP for 1995 = %. (Round your response to two decimal places.)

The growth rate of real GDP for 1996 = %. (Round your response to two decimal places.)

The growth rate of real GDP for 1997 = %. (Round your response to two decimal places.)

The growth rate of real GDP for 1998 = %. (Round your response to two decimal places.)

b. Calculate the average annual growth rate of real GDP for the period from 1994 to 1998.

Average annual growth rate = %. (Round your response to two decimal places.)

3. Capital can be differentiated between physical capital and human capital. Human capital is

- ☐ A. the goods manufactured to produce other goods and services.
- ☐ B. the new developments in machinery and software.
- ☒ C. the knowledge and skills workers acquire from education and training or from their life experiences
- ☐ D. the energy use to bring together the factors of production to produce goods and services.

Which of the following is the best example of human capital?

- ☒ A. a worker gets a college degree
- ☐ B. a manager buys workers more personal computers
- ☐ C. a manager devotes more spending to research and development
- ☐ D. a worker pays less taxes

4. Long-run growth in GDP is determined by

- ☒ A. capital, labor productivity, and technology.
- ☐ B. labor productivity, capital, and government expenditures.
- ☐ C. technology, investment, and consumption
- ☐ D. consumption, investment, and government expenditures.

Technological progress is affected by (*check all that apply*)

- ☒ A. new software developments
 - ☒ B. private property rights
 - ☒ C. entrepreneurship
 - ☒ D. investment in capital
 - ☐ E. population growth
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5. Which of the following is most likely to lead to sustained long-run growth?

- ☒ A. increases in labor productivity
 - ☐ B. transfer of workers from agricultural to industrial sectors
 - ☐ C. increases in the labor participation rate
 - ☐ D. exploitation of natural resources
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6. Potential GDP

- ☐ A. measures the maximum that a firm is capable of producing.
 - ☐ B. increases over time as the labor force grows.
 - ☐ C. increases over time as technological change occurs.
 - ☐ D. All of the above.
 - ☒ E. B and C only.
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7. The total value of saving in the economy must equal the total value of investment. Assume a closed economy, where:

I = Investment,
 $S = S_{\text{private}} + S_{\text{public}}$,
 S_{private} = Private Saving,
 S_{public} = Public Saving,
C = Consumption Expenditure,
G = Government Expenditure ,
Y = GDP,
TR = Government Tranfers.

Which one of the following expressions shows the investment-saving equality?

- ☐ A. $S = Y + T - TR - G$
 - ☐ B. $I = Y + TR - C - T$
 - ☐ C. $I = Y + TR - C - G$
 - ☒ D. $S = Y - C - G$
-

8. Which of the following is not a "loanable fund"?

- ☒ A. Real estate.
- ☐ B. Bonds.
- ☐ C. Mutual fund shares.
- ☐ D. Bank certificates of deposit.

Households supply loanable funds because of the

- ☐ A. wage income earned from working in the financial markets.
- ☐ B. rent income they receive as resource owner.
- ☐ C. profit income earned from running a money-lending business.
- ☒ D. interest income received from the borrowers.

Businesses demand loanable funds because

- ☒ A. firms need to borrow funds for new projects, such as building new factories or carrying out new research projects.
- ☐ B. firms need to borrow funds so that they can pay the wage costs and other recurring expenses of the business.
- ☐ C. loanable fund interest rates are always lower than the rate of return on their new investments.
- ☐ D. households charge a much higher rate of interest than the going rate of interest in the loanable funds market.

9. Consider the following data for a closed economy:

Y = \$15 trillion
C = \$7 trillion
I = \$2 trillion
TR = \$1 trillion
T = \$4 trillion

Use these data to calculate **a.** private saving, **b.** public saving, **c.** government purchases, and **d.** the government budget balance.

a. Private saving: \$ trillion. (Round your response to the nearest trillion dollars.)

b. Public saving: \$ trillion. (Round your response to the nearest trillion dollars.)

c. Government purchases: \$ trillion. (Round your response to the nearest trillion dollars.)

d. The government budget balance is \$ trillion and as a result the government budget is in deficit .
(Round your response to the nearest trillion dollars.)

10. Consider the following data for a closed economy.

Y = \$12 trillion
C = \$8 trillion
G = \$2 trillion
 $S_{\text{public}} = -\$0.5$ trillion
T = \$2 trillion

Now suppose that government purchases increase from \$2 trillion to \$2.7 trillion. If the values for Y and C are unchanged, what must happen to the values of S and I? Briefly explain.

The value of S must decrease by \$0.70 trillion and the value of I must decrease by \$0.70 trillion because $S = I$ and $I = Y - C - G$.

11. In a closed economy,

$Y = \$15$ billion

$C = \$12.0$ billion

$I = \$1.5$ billion

$TR = \$1.0$ billion

$T = \$3.0$ billion

Calculate each of the following (round each answer to one decimal place).

Private savings = \$ billion

Public savings = \$ billion

Total savings in this economy = \$ billion

The government's budget deficit or surplus = \$ billion

(enter a negative number for a deficit, positive number for surplus).

12.

Suppose the economy is currently in a recession and that economic forecasts indicate that the economy will soon enter an expansion.

As a result,

- ☐ A. expected profitability of new investment in plant and equipment increases and the demand for loanable funds falls.
- ☒ B. expected profitability of new investment in plant and equipment increases and the demand for loanable funds rises.
- ☐ C. expected profitability of new investment in plant and equipment increases and both the demand for and supply of loanable funds increase.
- ☐ D. expected profitability of new investment in plant and equipment increases and the supply of loanable funds rises, while demand remains unchanged.

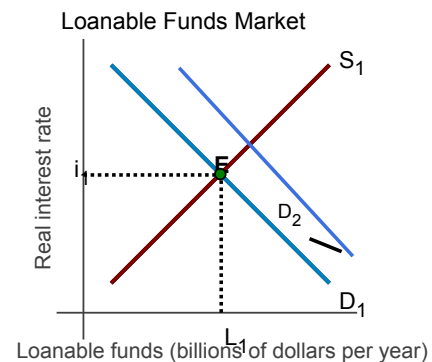
The initial loanable funds market is described by the demand for loanable funds curve, D_1 , and the supply of loanable funds curve, S_1 .

Using the line drawing tool to draw the demand or the supply curve if the economic forecasts indicate that the economy, which is currently in recession, will soon enter an expansion. Assume that the borrowers and lenders believe the forecast is accurate. Properly label each line.

Carefully follow the instructions above, and only draw the required objects.

Use the diagram to the right to answer the following question.

When borrowers and lenders believe that the forecast of economic expansion is accurate, the equilibrium real interest rate increases, the equilibrium quantity of loanable funds increases, and the quantity of saving and investment increases.



13. Firms care about their after-tax rate of return on investment projects.

The graph on the right represents the market for loanable¹ funds. The initial demand and supply curves are D_1 and S_1 , respectively.

Use the line drawing tool to shift the demand and or supply curve to illustrate the effect of an increase in business taxes. (For simplicity, assume no change in the federal budget deficit or budget surplus). Properly label this new line.

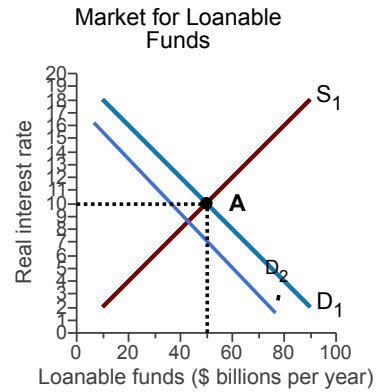
Carefully follow the instructions above, and only draw the required objects.

Use the graph to the right to answer the following questions.

When business taxes increase, the equilibrium interest rate decreases and the equilibrium quantity of loanable funds decreases.

What will be the effect of an increase in business taxes on the quantity of investment by firms and the economy's capital stock in the future?

- ☐ A. The quantity of investment will increase and the economy's future capital stock will decrease.
- ☒ B. The quantity of investment will decrease and the economy's future capital stock will decrease.
- ☐ C. The quantity of investment will decrease and the economy's future capital stock will increase.
- ☐ D. The quantity of investment will increase and the economy's future capital stock will increase.



1: Definition

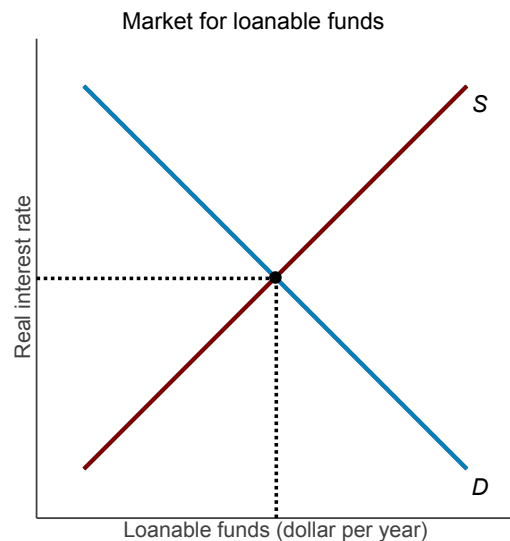
Market for loanable funds The interaction of borrowers and lenders that determines the market interest rate and the quantity of loanable funds exchanged.

14. Evaluate the following statement:

"Saving money is not lending. How can it be? When I save my money, I put it in a bank. I don't loan it out to someone else."

The statement is

- ☐ A. incorrect. The supply of loanable funds is determined by firms' willingness to borrow.
- ☐ B. correct. Depositing money in a bank is borrowing, not saving.
- ☐ C. correct. Depositing money in a bank is neither saving nor borrowing.
- ☒ D. incorrect. The supply of loanable funds is determined by household saving.



15. What are the names of the following events in a business cycle?

a. The high point of economic activity is called

- ☐ A. a zenith.
- ☐ B. a trough.
- ☐ C. a high.
- ☒ D. a peak.

b. The low point of economic activity is called

- ☐ A. a peak.
- ☐ B. a low.
- ☒ C. a trough.
- ☐ D. a bottom.

c. The period between the high point of economic activity and the following low point is called

- ☐ A. inflation.
- ☒ B. a recession.
- ☐ C. stagflation.
- ☐ D. an expansion.

d. The period between the low point of economic activity and the following high point is called

- ☐ A. stagflation.
- ☐ B. inflation.
- ☒ C. an expansion.
- ☐ D. an acceleration.

16. At the peak of a business cycle, which of the following is most likely to be true?

- ☐ A. The price level decreases.
- ☐ B. Output decreases, and employment falls.
- ☒ C. Interest rates are rising, and wages are rising faster than prices.
- ☐ D. Nominal wages and prices remain constant.

17. When tax revenue exceeds government spending (government purchases and transfer payments) there is a budget surplus .

When tax revenue equals government spending there is a balanced budget .

When tax revenue is less than government spending there is a budget deficit .

18. The financial system—either financial markets or financial intermediaries—provides savers and borrowers with all of the following except:

- ☒ A. The financial system provides security to savers by warranting that their funds are fully insured against loss.
- ☐ B. The financial system provides liquidity to savers by giving them the opportunity to buy and sell their financial securities.
- ☐ C. The financial system provides savers with facts and information about borrowers and about expected returns on their financial investments.
- ☐ D. The financial system provides risk sharing to savers by giving them the opportunity to diversify their funds among different investment choices.

